

Part 7

The branch tooling: one scene, two exports

The TF3 Export Prep add-on turns the TSC-structured scene into the FBX set the Model Editor wants, on a temporary copy, in one click. This part documents it, the naming map it applies, the texture-packing helper, and further scripts worth writing.

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Files: `tools/blender/tf3_export_prep.py` (Blender 4.2+ add-on, shares the TSC sidebar tab), `tools/tf3_pack_mga.py` (Python 3 + Pillow). The add-on was exercised headless in Blender 4.2 on a synthetic locomotive and its FBX re-imported to verify names, hierarchy and orientation; it has not been run through the real Model Editor here.

1 · What the add-on does and does not touch

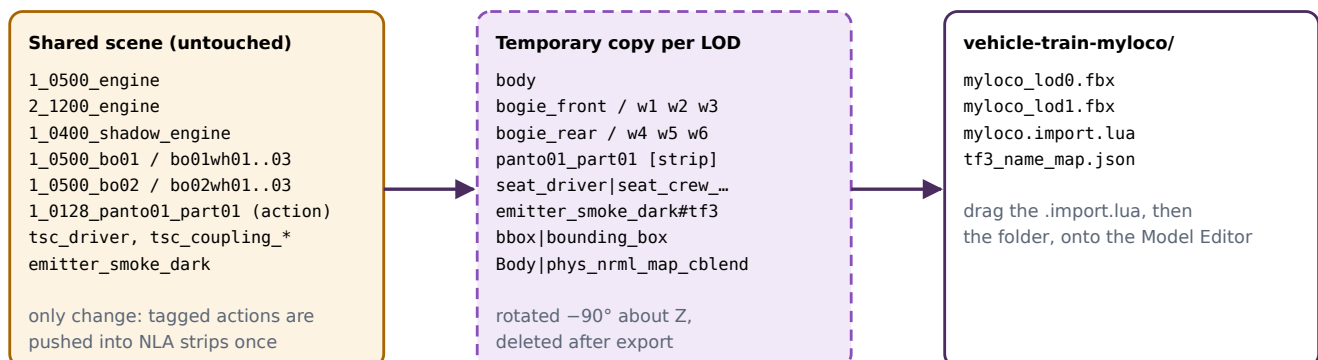


Figure 7.1 · Three stages of one click. Shadow meshes, number digits, light glow meshes, cab objects and coupler empties are skipped by default.

2 · Installing and the panel

- 1 Edit ► Preferences ► Add-ons ► Install from Disk... ► `tools/blender/tf3_export_prep.py` . It appears in the **TSC** sidebar tab as the **TF3 branch** panel (install the TSC Toolkit too; its Validate is the pre-flight for both targets).
- 2 Set **Export root** (a scratch folder or the mod's staging folder), **TF3 folder path** (`vehicle-train` OR `vehicle-wagon`) and **Model name**.
- 3 Fill the **Metadata seed**: engine type, power kW, tractive effort kN, top speed m/s, empty weight kg, years, reversible. These only populate the template; edit later in the Model Editor.
- 4 **Preview TF3 names** writes a text block showing every object's TF3 name per LOD and the skipped list. Fix tags until it reads right.
- 5 **Export TF3 FBX set**.

3 · Tags it reads

Custom property	On	Effect
tf3_name	Object	Use this node name instead of the automatic mapping.
tf3_event	Object	Push the object's action into an NLA strip of this name (<code>wheels</code> , <code>drive</code> , <code>forever</code> , <code>open_doors_left</code> , <code>front_forward_parts_on</code> ...). Done once; the TSC IA export still finds the animation via the strip.
tf3_skip	Object	Exclude from TF3 (cab pick meshes, TSC-only detail).
tf3_material	Material	Shortcode appended as <code>Name shortcode</code> so the importer picks that material type.
Empty <code>tsc_driver</code>	-	Becomes <code>seat_driver seat_crew_driving_upright</code> .

Custom property	On	Effect
Name contains emitter	Empty	Kept as a particle emitter locator (TF3 reads it by name).

The TSC IA export is unaffected by NLA strips

BRIAGE exports “the animation working in the Blender 3D view” (BRIAGE §8.5.1); an action in an NLA strip still plays in the viewport, so your `.ia` exports keep working after the push. If you prefer to keep actions active, duplicate the object into a TF3-only collection and tag only the duplicate.

4 ▪ What happens on export (step by step)

- 1 Collect LOD levels from object names (`1_` , `2_` ...; no prefix = LOD 1).
- 2 Push tagged actions into NLA strips named after their event (skipped if already present).
- 3 For each LOD: duplicate the LOD's meshes, plus empties and lights (so seats and emitters exist in every LOD), into a temporary collection; LOD-n objects take the place of their LOD-1 twins in the hierarchy.
- 4 Rename: wheels `w1...wN` ordered front to back by position; bogies `bogie_front / bogie_rear` ; body `body` ; others lose the prefix. If a name is already taken in the file, the copy is named `name#tf3` (the importer drops `#...`). Mesh datablocks are renamed the same so `.msh` files get clean names.
- 5 Materials: one copy per original material renamed `Name|shortcode` when `tf3_material` is set.
- 6 Rotate every root copy -90° about Z so +Y becomes +X. Children follow.
- 7 LOD 0 only: add a `bbox|bounding_box` cube around the meshes.
- 8 Export FBX: selection only, X forward / Z up, scale applied, mesh + empty + light, NLA strips baked (LOD 0 only), no leaf bones, textures not embedded.
- 9 Delete the temporary collection and its data; write `<name>.import.lua` and `tf3_name_map.json` ; log to the text block **TF3_Export**.

5 • The .import.lua it writes

```
fbxLoader = {
  predictMaterial = function(params) return nil end,          -- shortcodes in material names
  decide
  overrideProperties = function(params) return params.properties end,
  getMetadataMap = function()
    return {
      availability = { yearFrom = 1950, yearTo = 0 },
      description = { name = _("VEHICLE_SAMPLE_ELECTRIC_NAME"), description =
_("VEHICLE_SAMPLE_ELECTRIC_DESCRIPTION") },
      emissions = { noise = { score = -1 }, pollution = { score = -1 } },
      landVehicle = { engines = { { power = 6600.0, tractiveEffort = 300.0, type = "ELECTRIC" }
},
      topSpeed = 64.8, weightEmpty = 84000.0, weightMaxPayload = 0, friction =
0.02, brakeDeceleration = 2.5 },
      railVehicle = { config = { axles = { "w1", "w2", "w3", "w4", "w5", "w6" }, fakeBogies = { {
}, { } } } },
      transportVehicle = { carrier = "RAIL", transportModes = { "TRAIN", "ELECTRIC_TRAIN" },
engineTransportModes = { "ELECTRIC_TRAIN" },
      compartmentsList = { }, loadSpeed = -1, comfortFactor = -1,
arrivalDelay = 2000, departureDelay = 2000,
      reversible = false, filterTags = { "default" } },
      seatProvider = { drivingLicense = "RAIL", crewModels = { }, seats = { } },
      soundConfig = { soundSet = { name = "::/vehicle/train/shared/sound/train_electric.snd" } },
    }
  end,
}
```

Generated from the test scene. The three-function shape is the manual's [manual](#); the sound set path and `cost` field are placeholders [check](#). The axle list comes straight from the name map, so it cannot disagree with the FBX.

6 • The texture channel packer

```
python tools/tf3_pack_mga.py --metal body_metal.png --gloss body_gloss.png --ao body_ao.png --out
body_mga.tga
python tools/tf3_pack_mga.py --from-tsc-spec body_spec.png --ao body_ao.png --out body_mga.tga
# spec → gloss, metal = 0
```

Packs greyscale maps into the R (metal), G (gloss), B (AO) channels of a `_mga` texture [check](#) channel order against the `.mtl` page. With `--from-tsc-spec` it uses a TSC specular map as the gloss channel and a constant metalness. Requires Pillow (`pip install pillow`).

7 • Test results

Headless Blender 4.2, synthetic loco built to the Kuju sample's dimensions (body, two three-axle bogies, LOD-2 body, shadow mesh, pantograph with an action tagged `forward_parts_on`, driver empty, smoke emitter empty, coupler empty):

```

LOD 1 -> _lod0.fbx
  1_0500_engine -> body          1_0500_bo01 -> bogie_front      1_0500_bo01wh01 -> w1 ...
1_0500_bo02wh03 -> w6
  1_0128_panto01_part01 -> panto01_part01 [event forward_parts_on]
  tsc_driver -> seat_driver|seat_crew_driving_upright      emitter_smoke_dark ->
emitter_smoke_dark
LOD 2 -> _lod1.fbx      2_1200_engine -> body (+ empties)
skipped (TSC-only): 1_0400_shadow_engine
re-imported _lod0.fbx:  body (0,0,0) · bogie_front (5.58, 0, 0.27) · bogie_rear (-5.58, 0, 0.27)
                        w1 (7.80, 0, 0.52) ... w6 (-7.79, 0, 0.52), parented to their bogies
                        seat at (6.90, 0.40, 0.86) · materials all "Body|phys_nrml_map_cblend"
scene after export: unchanged object list; pantograph action now in NLA track "forward_parts_on"

```

Positions confirm the rotation (TSC +Y 5.583 → TF3 +X 5.58), the front-to-back axle numbering and the preserved parenting.

8 • Further scripts worth building (concepts)

Idea	What it would do	Friction removed
Metadata from the TSC blueprint	Parse the TSC engine/sim XML (the <code>tsc_crosscheck</code> code already does) and fill the <code>.import.lua</code> seed: power, TE, speed, mass, engine type, years. Zero re-typing.	Two physics forms
Parts-event instancer	For objects tagged <code>tf3_parts = "front"</code> , create raised/lowered or lit/unlit duplicates with the right <code>...parts_on/off</code> one-key animations automatically.	Hand-made on/off pairs
Seat row generator	Place <code> seat_sitting</code> empties on a grid inside a coach from pitch and width parameters.	Dozens of empties
Label placer	Make the two nested <code> label</code> empties from a selected rectangle face.	Fiddly locator pairs
Model Editor bulk file	Write the <code>Scene;/ModRefName;/Path;</code> text file for a whole train set so one drag imports everything.	Many drags
.mdl diff	After import, parse the saved <code>.mdl</code> (Lua table) and compare node names/transforms against <code>tf3_name_map.json</code> and the scene, like <code>tsc_crosscheck</code> does for TSC.	"What did the importer do?"
Texture converter	TSC diffuse + spec → <code>_albo</code> , <code>_mga</code> , flat <code>_nrm</code> , blank <code>_cdr</code> with a livery mask stub.	PBR conversion

9 • The daily loop with both targets

- 1 Model and animate in the shared scene with TSC names; add the three TF3 habits (axles as objects, driver empty, independent actions).
- 2 TSC toolkit **Validate**.
- 3 TSC: BRIAGE IGS/IA → Measure + Patch → BPE2 → crosscheck → game (TSC guide Part 7).

- 4 TF3: **Preview TF3 names** → **Export TF3 FBX set** → drag `.import.lua` , drag folder → Model Editor validate → preview → game (this guide Part 6).
- 5 A change to the model goes through both lanes; a change to physics or metadata stays in its own lane.